Use Knowledge Bases and HMM Approach to Predict Protein Secondary Structure

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ABSTRACT

In this paper, knowledge bases and HMM approach are used to predict protein secondary structure. This system contains the databases of primary, secondary and tertiary structures of proteins. PSI-BLAST is used to find all the domain profiles of proteins in EC2. Then these similar domains are grouped and built into domain profile databases.

Next, the Hidden Markov Model's (HMM) parameters can be trained by using the data of the proteins with recognized structure. The 2D structure of the query protein's domain can be identified, if it's matched homology profile in the databases is found. Then the 2D structures of remaining fragments can be predicted by using HMM approach. The accuracy of this method is above 80% for enzyme protein.

1 INTRODUCTION

Functional characterization of a protein sequence is the most frequent problem in biology. This task is usually facilitated with accurate three-dimensional (3D) structure of the studied protein. The structure of protein gives important information, such as the location of probable function (or interaction sites), identification of distantly related proteins, and important regions of the protein sequence that are involved in maintaining the structure. However, it is difficult and time consuming to experimentally determine protein structure. For these reasons, protein structure predicting methods [1,2] have received a lot of attention.

Predicting the secondary structure of a protein from its sequence is a fundamental step in predicting the overall quaternary structure of a protein. Secondary structure prediction [3] is often limited to three components: α-helix, extended β-sheet, and coil. These forms are further folded together to form the tertiary, or globular, structure of a protein, and the quaternary structure is the result of the interaction between independently folded tertiary units.

Many methods have been proposed for protein secondary structure prediction. They can be categorized into four broad groups. The first group contains statistical based methods that make use of parameters obtained from analysis of known sequences and 3D structures [4, 5]. The predictive accuracy of this group of methods stays around 50 - 70%. The second group is "nearest neighbor" methods [6, 7].

These methods use sequence alignments to find sequences of known structure that have similar segments to the query sequence. The predictive accuracy of this group of methods stays 68 -75%. The third group of methods is based on machine learning algorithms (HMM [8,9] and neural network models [10, 11]). The performance is in the range of 64 –75%. The fourth group of methods [12] combines previous methods, and its accuracy can reach to 72.9%.

In this paper, we develop a system that contains the databases of primary sequence, secondary and tertiary structure of protein. These databases are converted from Brookhaven's Protein Data Bank (PDB) [13, 14, 15]. Then, the Hidden Markov Model (HMM) [16, 17] is set up by statistically evaluating parameters from the proteins with recognized secondary structure. The forward, backward, and Baum-Welch algorithm [18, 19, 20] are used to train data, and Viterbi algorithm [21] is applied for finding optimal path. The accuracy of using HMM, Viterbi and to predict enzyme protein [22] is 64.34%.

To improve predicting accuracy, PSI-BLAST [23, 24, 25] is used to find 22,316 similar domain profiles between 1,025 enzyme proteins and 18,000 structure-recognized proteins. These domain profiles are further grouped into 2,926 groups with similarity ≧ 30%. These groups are stored into domain profile databases [26]. The 2D structure of the query protein's domain can be identified, if it's matched homology profile in the domain profile databases is found [27, 28, 29]. In final step, the 2D structures of remaining fragments can be predicted by using HMM approach. The accuracy of this method is above 80% for enzyme protein.

In the Section 2, HMM approach is described to predict 2D structure and it's performance is discussed also. Domain profile databases and HMM are used to increase the accuracy of prediction, and resulting performance is analyzed in Section3. Finally the conclusion is given in Section 4.

2 USING HMM APPROACH TO PREDICT

An HMM is established for predicting 2D structure in Section 2.1. ER model of protein structure database is built in Section 2.2 first. Then this database is used to train HMM data in Section 2.3. Viterbi's forward and backward algorithm and Baum-Welch algorithm are applied in this section for data training also. The resulting HMM parameters are evaluated in Section 2.4. Finally, the performance of this approach is given in Section 2.5.

2.1 Our Hidden Markov model:

Our Hidden Markov model is shown in Figure 2.1which are discussed below:

- Helix unit: represents the secondary structure of this amino acid is helix..
- Sheet unit: represents the secondary structure of this amino acid is sheet.
- Coil unit: represents the secondary structure of this amino acid is coil.
- P(ALA)..P(TYR) in each box: are the probability of amino acid that occurs in this type of secondary structure . They are also called emission probabilities.
- P(x1_x2): are transition probability from state x1 to state x2.

The emission probabilities and transition probabilities will be trained using forward, backward and Baum-Welch algorithm in Section 2.4. It is noted that sequence can starts from any secondary component (helix, Sheet or coil) in this design, and then Viterbi algorithm will be adopted to find the most possible path in Section.2.5

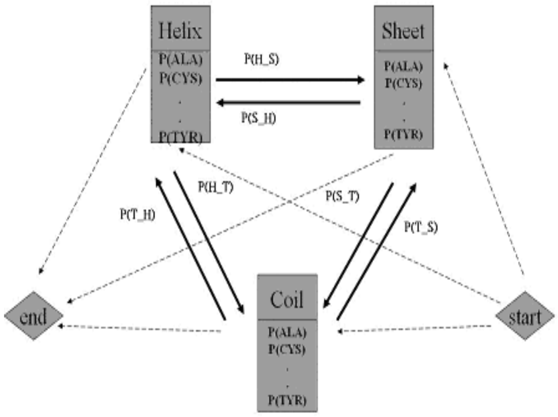

Figure 2.1 An Hidden Markov model for predicting 2D structure

<u>2.2 E/R Model of protein structure database</u>:

Figure 2.2 is the E/R model of our protein structure database. The database is build from 17945 protein data files.

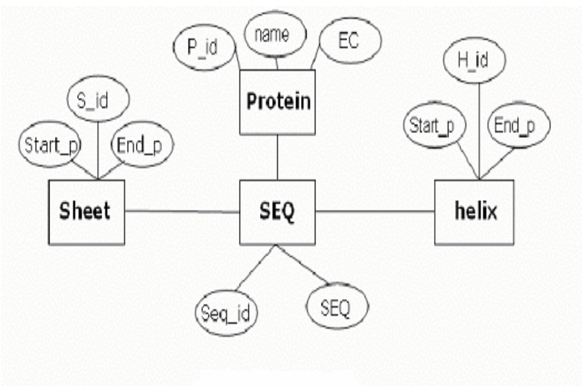

Figure 2.2 E/R Model for HMM training data.

<u>2.3 HMM Data Training:</u>

We choose the enzyme protein data in EC2 to train HMM parameters iteratively using Baum-Welch algorithm. The EC2 enzyme protein contains 2261 records in our database. The figure2.3 shows the flowchart of training HMM parameters.

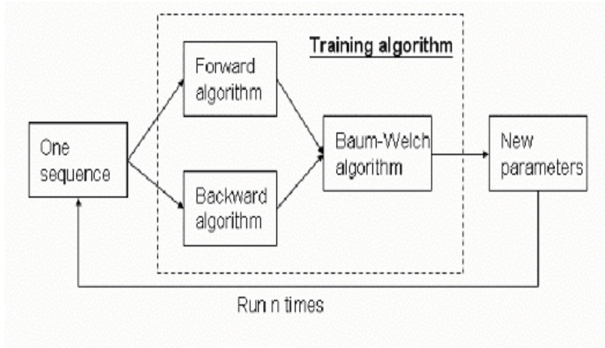

Figure2.3 HMM training process.

Let the variables are defined below:

A : A set of state transition probability A = $\{a_{ij}\}$.

B : is a emission probability set in each of the state B = $\{b_j(k)\}$.

a_{kl} : is transition probability from state k to state l.

$e_l(b)$: is the probability that symbol b is seen in state l.

x_i: is the i-th symbol in the sequence.

Θ: represents the entire current parameter values (all probabilities).

Baum-Welch algorithm can be applied to train HMM parameters. This algorithm is described as:

Initialization：Pick arbitrary model parameters.

Recurrence:

Set all the A and E variables to their pseudo-count values r(or to zero).

For each sequence j=1… n：

Calculate $f_k(i)$ for sequence j using the forward algorithm.

Initialtion（i=0）: $f_0(0)=1, f_k(0)=0 \quad K>0$

Recursion（i=1...L）: $f_l(i)=e_l(x_i)\sum_k f_k(i-1)a_{kl}$

Termination: $P(x)=\sum_k f_k(L)a_{k0}$

Calculate $b_k(i)$ for sequence j using the backward algorithm:

Initialtion (i=L) : $b_k(L) = a_{k0}$ for all k

Recursion (i=L...1) : $b_k(i) = \sum_l a_{kl} e_l(x_{i+1}) b_l(i+1)$

Termination : $P(x) = \sum_l a_{0l} e_l(x_1) b_l(1)$

Add the contribution of sequence j to A and E:

$$A_{kl} = \sum_j \frac{1}{P(x^j)} \sum_i f_k^j(i) a_{kl} e_l(x_{i+1}^j) b_l^j(i+1)$$

$$Ek(b) = \sum_j \frac{1}{P(x^j)} \sum_{\{i / x_i^j = b\}} f_k^j(i) b_k^j(i)$$

Calculate the new model parameters:

$$a_{kl} = \frac{A_{kl}}{\sum_{l'} A_{kl'}} \quad \text{and} \quad e_k(b) = \frac{E_k(b)}{\sum_{b'} E_k(b')}$$

Calculate the new log likelihood of the model.

$$\log P(x^1,\dots,x^n / \Theta) = \sum_{j=1}^{n} \log P(x^j | \Theta)$$

Termination：

Stop if the change in log likelihood is less than some predefined threshold or the maximum number of iterations is exceeded.

2.4 Result of HMM parameters:

After implementing the Baum-Welch algorithm, HMM parameters defined in Figure 2.2 can be trained completely. These parameters are listed in table 2.1.

Table 2.1 The resulting parameters of our HMM.

Data source : Enzyme_EC2			
2D Structure: HELIX			
Amino acid	P	Amino acid	P
H_A	0.1009	H_M	0.0273
H_C	0.0089	H_N	0.0352
H_D	0.0617	H_P	0.0238
H_E	0.0817	H_Q	0.0527
H_F	0.045	H_R	0.0506
H_G	0.0385	H_S	0.0456
H_H	0.0318	H_T	0.0439
H_I	0.0525	H_V	0.0547
H_K	0.0689	H_W	0.0185
H_L	0.1219	H_Y	0.0358

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H_K	0.0689	H_W	0.0185
H_L	0.1219	H_Y	0.0358

Data source : Enzyme_EC2			
2D Structure: TURN			
Amino acid	P	Amino acid	P
T_A	0.0685	T_M	0.0185
T_C	0.0124	T_N	0.0483
T_D	0.0742	T_P	0.0841
T_E	0.0564	T_Q	0.0292
T_F	0.0365	T_R	0.0497
T_G	0.1051	T_S	0.0562
T_H	0.0348	T_T	0.0532
T_I	0.0406	T_V	0.0524
T_K	0.0615	T_W	0.0144
T_L	0.0705	T_Y	0.0335

Transition	
Three state	
Transition	P
H_H	0.8982
H_S	0.0075
H_T	0.0943
S_H	0.018
S_S	0.8154
S_T	0.1667
T_H	0.0804
T_S	0.069
T_T	0.8506

2.5 Evaluating the performance of HMM approach:

We randomly choose 30 protein sequences in order to test system's performance. The resulting HMM parameters and Viterbi algorithm are used to predict 2D structure of protein, and the accuracy of every protein sequence are listed in Table2.2 .The most widely used way to determine prediction quality is defined in Q_3 :

$$Q3 = [(PH + PS + PT)/N] \times 100\%$$

where PH，PS and PT are the numbers of correctly predicted states for three secondary components. N is the total number of residues.

Table 2.2 The performance of HMM approach for enzyme proteins

Using HMM to predict Enzyme_EC2							
ID	Protein name	Chain	P	ID	Protein name	Chain	P
1	1OMH	A	0.587	16	1BKO	A	0.70576
2	1BHJ	A	0.611	17	1D2H	B	0.64981
3	1BJ4	A	0.628	18	1AQI	B	0.67663
4	1BWD	A	0.776	19	1BC5	A	0.7177
5	1C9Y	A	0.755	20	1EQ2	A	0.61553
6	1BQ2		0.576	21	1HNN	B	0.6212
7	1LXA		0.67	22	1HVY	A	0.75972
8	1ORT	J	0.553	23	1HW3	A	0.47591
9	1VZD		0.679	24	1NJE		0.52906
10	1ORT	C	0.702	25	1VZE		0.59272
11	1XVA	A	0.56	26	1QD1	A	0.6659
12	6JDW	A	0.643	27	1C9Y	A	0.6502
13	1D2C	B	0.716	28	1HZW	B	0.63242
14	1BHJ	D	0.685	29	1JKX	D	0.68297
15	1VID		0.69031	30	1KHH	B	0.49553
Total Accuracy		64.34%		Total PDB		30	

3 IMPROVING OUR PREDICTION METHOD

The concept of our prediction method is improved in Section 3.1. PSI-BLAST is used to search entire similar domain profiles in Section 3.2. These profiles are merged and stored into profile database in Section 3.3. Prediction performance is evaluated in Section 3.4 for EC2 proteins, and is evaluated in Section 3.5 for all structure recognized proteins.

3.1 The concept of our approach:

Although the EC2 group has 1633 protein records, many records are identical to others. So we reduce the unique protein records to 1025. Then we use PSI-BLAST to search similar domain profiles for each enzyme protein of EC2. Figure 3.1 shows the flowchart of predicting 2D structure.

There are two steps for finding domain profiles. The first step is to find the relative domain sequences of query protein in Gene Bank using PSI-BLAST. The second step is to search the corresponding group in domain profile databases. After that, get the most similar domain within this group using global alignment algorithm, and corresponding 2D structures of these domains can be retrieved from domain profile databases. The hidden markov model built in Section 2 can be applied to predict the remaining fragments.

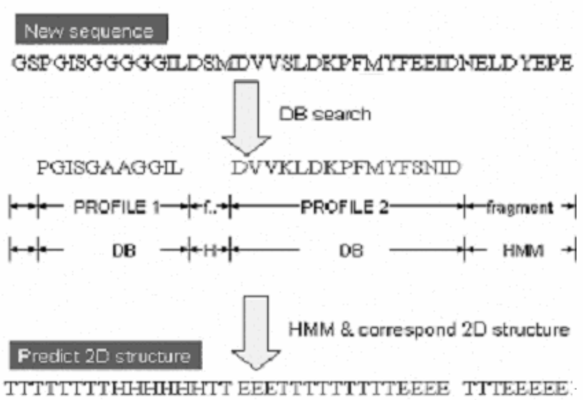

Figure 3.1 The Flowchart of Prediction 2D Structure

3.2 Search similar domains using PSI-BLAST:

The principal design goals in developing the Position-Specific Iterated BLAST (PSI-BLAST) were speed, simplicity and automatic operation. It is shown in Figure 3.2. The procedure of PSI-BLAST can be summarized in five steps:

(1) PSI-BLAST takes a single protein sequence as an input and compares it to protein database, using the gapped BLAST program.

(2) This program constructs a multiple alignment and generates a profile, if any significant local alignment is found. The original query sequence serves as a template for the multiple alignment and profile. Different numbers of sequences can be aligned in different template positions.

(3) The profile is compared to the protein database in order to seek local alignments. After a few minor modifications, the BLAST algorithm can corrects and uses this profile directly.

(4) PSI-BLAST estimates the statistical significance of the local alignments, using profile substitution scores that are related to gap scores. The statistical theory and parameters for gapped BLAST alignments still remain applicable to profile alignments.

(5) Finally, PSI-BLAST iterates, by returning to Step (2), an arbitrary number of times or until convergence.

Profile-alignment statistics allow PSI-BLAST to proceed as a natural extension of BLAST; the results produced in iterative search steps are comparable to those data produced from the first pass. Unlike most profile-based search methods, PSI-BLAST runs as one program, starting with a single protein sequence, and the intermediate steps of multiple alignment and profile construction are invisible to the user.

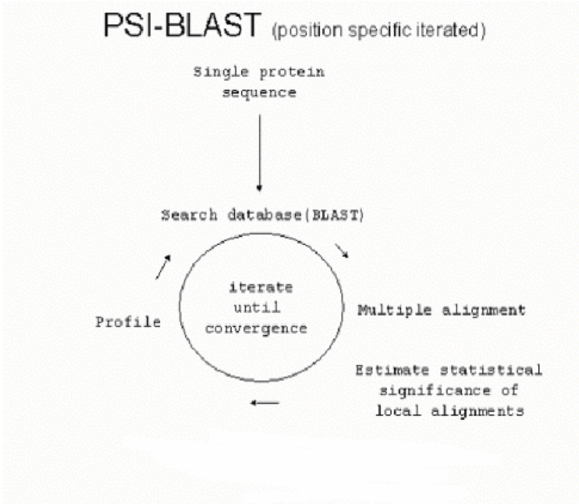

Figure 3.2 Search related profiles using PSI-BLAST.

3.3 Merge similar profiles:

There are 22316 profiles found from 1025 EC2 proteins. We use global alignment algorithm to classify all profiles which are collected from PSI- BLAST program. Figure 3.3 is the process of producing domain profile databases. Each step is described below:

(1) In order to reduce time complexity, we only compare two sequences with length difference < 30% of maximum sequence length.

(2) We use global alignment algorithm to count identical codes between two sequences.

If the number of identical codes > 30% of sequence length, then merge these two profiles into one group. These data are represented as domain profile databases.

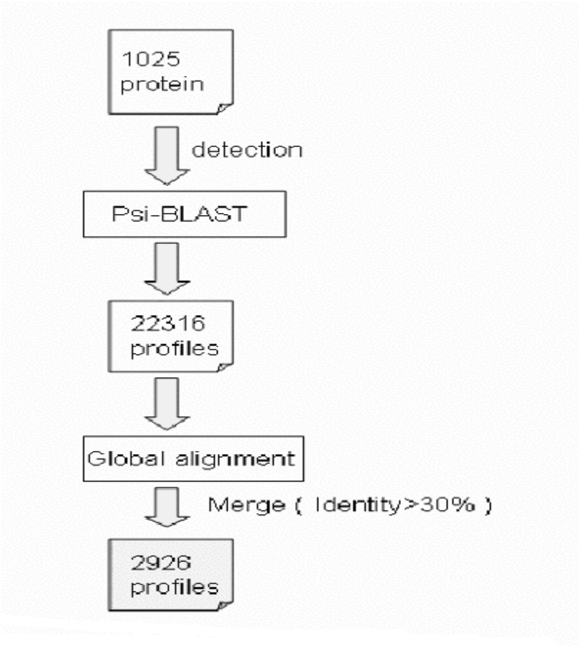

Figure 3.3 Collecting domain profile databases from PSI-BLAST files.

3.4 Evaluate prediction performance for enzyme proteins:

We choice 30 protein sequences from 1025 records of EC2 randomly, and use our approach to predict 2D structure. Evaluating procedure are given below:
(1) Use PSI- BLAST to detect subsequence of target sequence.
(2) Use global alignment to count the identical code number between target subsequence and every profile sequence. Then choose a best profile sequence which has maximum number of identical codes with target subsequence. The 2D structure of target subsequence will be substitute with this profile sequence structure.
(3) After that, several fragments may still remain between subsequences provided in (1) Then we can use HMM approach to predict 2D structure of remaining fragments.
(4) The accuracy of predicted structure can be evaluated by comparing with recognized
structure. The performance result for enzyme proteins is shown in Table 3.1.

Table 3.1 Evaluate the performance for enzyme proteins only.

Pdb_name	Chain	Accuracy(%)	Pdb_name	Chain	Accuracy(%)
2TPS	A	95.02	1JOW	A	91.10
1BHN		92.23	3THI	A	93.56
1VP3		91.65	2NMT		80.81
1RRB		57.94	1DOS		60.01
1NGS	A	94.56	1A25	A	89.26
1HQK	A	98.06	1FAQ		73.07
1HKB	A	94.41	1E2D		90.23
1K8R	A	80.12	1DTY	A	81.35
1E0H	B	65.90	1IB1	E	70.50
1DTY	A	81.35	1GCX		69.12
1DLG	A	91.17	1UBV		92.10
1C3Q	A	69.12	1I35		76.70
1BJN	A	94.44	1SGG		81.33
1BTK	A	84.62	2ECK	A	80.37
1JSU	C	73.81	1NAW	A	83.77
TOTAL ACCURACY		80.12	TOTAL PDB		30

3.5 Evaluate performance for all structure recognized proteins:

If we choice 30 protein sequences from 18,000 structure recognized proteins, and use our method to predict 2D structure. The result is evaluated in Table 3.2 below. The accuracy is 68.46%. It is noted that our domain profile databases are collected only from enzyme proteins. These databases can be extended by collecting all PSI-BLAST files of the structure recognized proteins. It is also suggested that all the structure recognized proteins are classified into different classes based on their function. Each class can has it’s own HMM parameters. The most possible class for a target protein can be predicted by finding the highest Viterbi probability from different sets of HMM parameters.

Table 3.2 Evaluate the performance for all structure recognized proteins.

Pdb_name	Chain	Accuracy(%)	Pdb_name	Chain	Accuracy(%)
1C9M	A	66.31	1HNI	A	67.81
1AZI		72.54	1GCW	B	70.12
1II6		62.92	1RDS		70.91
2MYC		68.63	1DQM	H	66.54
1QIB	A	70.12	1COY		53.65
1HQK	B	81.43	1VRE	A	59.62
1K62	A	63.65	1G1V	A	80.13
1FPS		74.23	1K96	A	71.16
1IEZ	A	79.44	1GGH	A	65.42
1BOF		62.21	1EGE	C	64.32
1E69		57.84	1I27		63.44
1JXK	A	65.94	1DAX		72.16
1NIB	B	76.67	1FF5	A	72.31
1JE3	A	71.58	1SNM		63.12
1TIU		69.81	1QKN	A	69.74
TOTAL ACCURACY		68.46	TOTAL PDB		30

4 CONCLUSION

Predicting the secondary structure of a protein using sequence is a fundamental step in predicting the overall (quaternary) structure of a protein. In this paper, we have developed a knowledge base system and HMM approach to predict protein 2D structure . To predict the structure of a target sequence, this sequence can get similar domain 2D structure from PSI-BLAST and our domain profile databases. The remaining fragments can be predicted by using HMM approach.. The average accuracy of our method is 80.12% for enzyme protein.

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